

## ENGINEERING MEMO

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| <b>To:</b>       | Plant Project Review Board                                                       |
| <b>From:</b>     | Process Engineering Review                                                       |
| <b>Date:</b>     | 28 June 2026                                                                     |
| <b>Subject:</b>  | CH <sub>4</sub> / C <sub>2</sub> H <sub>4</sub> splitter shortcut FUG validation |
| <b>Revision:</b> | A                                                                                |

### Modelling basis

The validation treats the service as a binary methane/ethene split at 7.0 bar. Relative volatility varies across the column, so the shortcut method uses an effective  $\alpha(\text{CH}_4/\text{C}_2\text{H}_4) = 1.85$ , obtained as the geometric mean of the distillate-end value (2.500) and the bottoms-end value (1.369), per the standard Fenske averaging convention. The feed is subcooled with  $q = 1.2$ . The kettle reboiler is included as an equilibrium stage; the total condenser is excluded from the stage count. Results are shortcut estimates and do not include hydraulic capacity, tray/packing efficiency, or rigorous thermodynamic regression.

### Overhead-GC interpretation

The analyser is specified as a vapor-line GC above the total condenser, so the reported CH<sub>4</sub>:C<sub>2</sub>H<sub>4</sub> vapor peak ratio of 18.0:1.00 was normalized directly as the distillate composition. This gives  $x_{D,\text{CH}_4} = 0.9474$  and  $x_{D,\text{C}_2\text{H}_4} = 0.0526$ ; the bottoms specification is  $x_{B,\text{CH}_4} = 0.0100$  and  $x_{B,\text{C}_2\text{H}_4} = 0.9900$ .

### Shortcut results and balance check

| Result                                                  | Value           | Comment                                                                             |
|---------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|
| Effective relative volatility ( $\alpha_{\text{eff}}$ ) | 1.8500          | Geometric mean of 2.500 (top) and 1.369 (bottom)                                    |
| Underwood root theta                                    | 1.1243          | Root selected between $\alpha_{\text{HK}} = 1.000$ and $\alpha_{\text{eff}} = 1.85$ |
| Minimum reflux ratio $R_{\text{min}}$                   | 0.9918          | Back-substitution gives $R_{\text{min}} + 1 = 1.9918$                               |
| Operating reflux ratio $R$                              | 1.2397          | Specified as $1.25 \times R_{\text{min}}$                                           |
| $N_{\text{min}}$                                        | 12.17           | Includes kettle reboiler; excludes total condenser                                  |
| Gilliland/Eduljee $N$                                   | 27.29           | Continuous theoretical stage estimate at operating reflux                           |
| D/F and B/F                                             | 0.7793 / 0.2207 | Molar basis; $D/F + B/F = 1.0000$                                                   |

### Engineering limitations for board review

The shortcut package supports a consistent review basis, but final design lock should still check sensitivity to the effective relative volatility and  $q$ , confirm the GC/distillate interpretation, and compare the continuous theoretical stage requirement against the actual column stage count, efficiency basis, pressure profile, and hydraulic constraints.